9th Class • Punjab Board • PECTAA New Syllabus

Chapter 2: Kinematics

Complete solved exercise — vectors, motion graphs, equations of motion and free-fall numericals.

Chapter 2 • Complete official exercise • 10 MCQs + 8 short + 6 constructed + 7 comprehensive + 10 numericals

What this chapter is about

• **Rest and motion** are relative to a reference point. Distance is total path length; displacement is the straight change in position with direction.

• **Speed** is scalar and velocity is vector. Acceleration is the rate of change of velocity and can result from changing speed, direction or both.

• **Graphs:** distance–time gradient gives speed, displacement–time gradient gives velocity, speed–time gradient gives acceleration and area under a speed–time graph gives distance.

• **Uniformly accelerated motion:** $v = u + at$, $S = ut + \frac{1}{2}at^2$ and $v^2 - u^2 = 2aS$. For vertical motion, replace a by g with a clearly stated sign convention.

Exercise coverage: All official sections A–E are solved in book order, including graph interpretation, vector drawing and ten step-by-step numericals. Take $g = 10\ m\ s^{-2}$ as used by the textbook.

A. Multiple Choice Questions

1. The numerical ratio of displacement to distance is:

(a) always less than one (b) always equal to one
(c) always greater than one (d) equal to or less than one

Answer: (d) equal to or less than one — Displacement magnitude can equal distance on a straight path without reversal, but can never exceed the actual path length.

2. If a body does not change its position with respect to some fixed point, then it will be in a state of:

(a) rest (b) motion
(c) uniform motion (d) variable motion

Answer: (a) rest — Rest means position remains unchanged relative to the chosen reference point.

3. A ball is dropped from the top of a tower, the distance covered by it in the first second is:

(a) 5 m (b) 10 m
(c) 50 m (d) 100 m

Answer: (a) 5 m — From rest, $S = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 1^2 = 5$ m.

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4. A body accelerates from rest to a velocity of 144 km h^{-1} in 20 seconds. The distance covered by it is:

(a) 100 m (b) 400 m
(c) 1400 m (d) 1440 m

Answer: (b) 400 m — 144 km h^{-1} = 40 m s^{-1}. Average velocity = (0 + 40)/2 = 20 m s^{-1}, so S = 20 × 20 = 400 m.

5. A body is moving with constant acceleration starting from rest. It covers a distance S in 4 seconds. How much time does it take to cover one-fourth of this distance?

(a) 1 s (b) 2 s
(c) 4 s (d) 16 s

Answer: (b) 2 s — From rest $S \propto t^2$. For S/4, t^2 becomes one-fourth, so time becomes one-half: 4/2 = 2 s.

6. The displacement time graphs of two objects A and B are shown in the figure. Point out the true statement from the following:

(a) The velocity of A is greater than B. (b) The velocity of A is less than B.
(c) The velocity of A is equal to that of B. (d) The graph gives no information in this regard.

Answer: (a) The velocity of A is greater than B. — A has a positive displacement–time gradient, while B is horizontal and has zero velocity; therefore A has greater velocity.

7. The area under the speed-time graph is numerically equal to:

(a) velocity (b) uniform velocity
(c) acceleration (d) distance covered

Answer: (d) distance covered — Each small rectangle has area speed × time, which is distance; their sum is total distance.

8. Gradient of the speed-time graph is equal to:

(a) speed (b) velocity
(c) acceleration (d) distance covered

Answer: (c) acceleration — Gradient = change in speed ÷ change in time = acceleration.

9. Gradient of the distance-time graph is equal to the:

(a) speed (b) velocity
(c) distance covered (d) acceleration

Answer: (a) speed — For a distance–time graph, gradient = distance travelled ÷ elapsed time = speed.

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10. A car accelerates uniformly from 80.5 km h^{-1} at t = 0 to 113 km h^{-1} at t = 9 s. Which graph best describes the motion of the car?

(a) (a) Curved upward graph
(b) (b) Straight line through the origin
(c) (c) Straight rising line with positive velocity intercept
(d) (d) Straight falling line

Answer: (c) (c) Straight rising line with positive velocity intercept — Uniform acceleration gives a straight velocity–time line; the initial velocity is nonzero, so the line has a positive intercept and positive gradient.

B. Short Answer Questions

2.1. Define scalar and vector quantities.

A **scalar quantity** is described completely by magnitude and unit alone, such as mass or time. A **vector quantity** requires magnitude, unit and direction, such as displacement or force.

2.2. Give 5 examples each for scalar and vector quantities.

Scalar quantities	Vector quantities
distance, speed, mass, time, temperature	displacement, velocity, acceleration, force, momentum

2.3. State head-to-tail rule for addition of vectors.

Draw the first vector, then draw the second with its tail at the head of the first, keeping magnitude and direction unchanged. Continue for other vectors. The resultant is the arrow from the tail of the first vector to the head of the last.

2.4. What are distance-time graph and speed-time graph?

A **distance–time graph** plots distance travelled on the vertical axis against time horizontally; its gradient is speed. A **speed–time graph** plots speed vertically against time; its gradient is acceleration and the area beneath it is distance travelled.

2.5. Falling objects near the Earth have the same constant acceleration. Does this imply that a heavier object will fall faster than a lighter object?

No. Neglecting air resistance, all freely falling objects have the same acceleration g regardless of mass, so objects dropped together from the same height fall equally fast. In air, shape and air resistance can make their motions differ.

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2.6. The vector quantities are sometimes written in scalar notation (not bold face). How is the direction indicated?

When bold type is not used, place a **small arrow over the symbol**, such as $\vec{v}$ or $\vec{F}$. In one dimension, a declared positive direction and plus/minus signs can also indicate direction.

2.7. A body is moving with uniform speed. Will its velocity be uniform? Give reason.

Not necessarily. Velocity is uniform only if both speed and direction remain constant. A body moving at uniform speed around a circle continuously changes direction, so its velocity changes and it accelerates.

2.8. Is it possible for a body to have acceleration? When moving with: (i) constant velocity (ii) constant speed

(i) **Constant velocity:** No; acceleration is zero because neither speed nor direction changes. (ii) **Constant speed:** Yes; acceleration is possible if direction changes, as in uniform circular motion.

C. Constructed Response Questions

2.1. Distance and displacement may or may not be equal in magnitude. Explain this statement.

They are equal when a body travels in a straight line from start to finish without reversing direction. If the path curves or the body reverses, total distance is greater than displacement magnitude. On a complete round trip, distance is nonzero but displacement is zero.

2.2. The muzzle velocity of one gun with a longer barrel is lesser than that of another gun with a shorter barrel. In which gun is the acceleration of the bullet larger? Explain your answer.

The bullet in the **shorter barrel** has the larger acceleration under the stated conditions. Starting approximately from rest, $v^2 = 2aS$, so $a = v^2/(2S)$. The shorter barrel has both a larger muzzle velocity v and a smaller travel distance S; both effects increase a.

2.3. For a complete trip, average velocity was calculated. Its value came out to be positive. Is it possible that its instantaneous velocity at any time during the trip had the negative value? Give justification of your answer.

Yes. Average velocity depends only on net displacement and total time, whereas instantaneous velocity can change direction. For example, a traveller may move +10 m and then −4 m; velocity is negative during the return portion, but net displacement is +6 m and average velocity is positive.

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2.4. A ball is thrown vertically upward with velocity v. It returns to the ground in time T. Which of the following graphs correctly represents the motion? Explain your reasoning.

Graph (c) is correct when upward is chosen positive. Velocity starts at +v and decreases uniformly with gradient −g. At T/2 it is zero at the highest point; afterwards it becomes negative and reaches −v on returning to the launch level.

$$v(t) = v - gt$$

At highest point: $t = T/2$ and $v = 0$

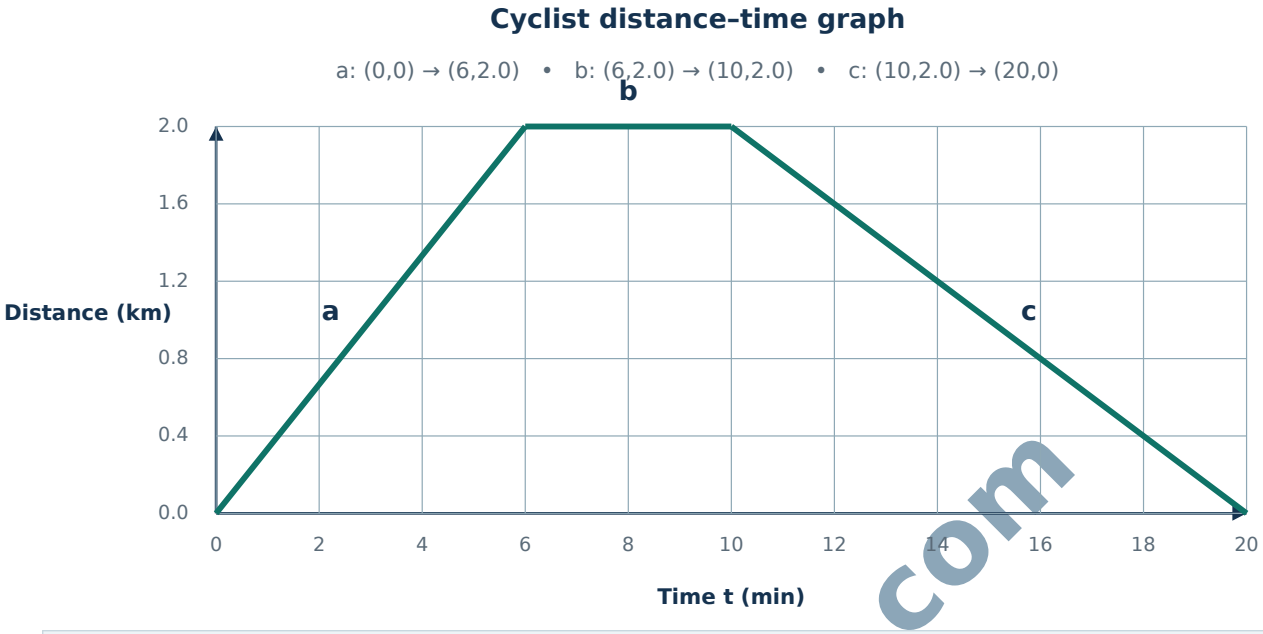
On return: $t = T$ and velocity $= -v$

A V-shaped speed graph would describe |velocity|, but the exercise axis is signed velocity v, so the straight line must cross zero and continue below the time axis.

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2.5. The figure given below shows the distance-time graph for the travel of a cyclist. Find the velocities for the segments a, b and c.

Figure 1 — Redrawn graph with coordinates used for each gradient



Segment a: $v = (2.0 - 0)/(6 - 0) = +0.333$ km min^{-1} $= +5.56$ m s^{-1}

Segment b: $v = (2.0 - 2.0)/(10 - 6) = 0$

Segment c: $v = (0 - 2.0)/(20 - 10) = -0.200$ km min^{-1} $= -3.33$ m s^{-1}

Segment b is rest. Segment c is the return journey; its speed is 0.200 km min^{-1} (12 km h^{-1}), while its signed velocity is negative because displacement decreases.

2.6. Is it possible that the velocity of an object is zero at an instant of time, but its acceleration is not zero? If yes, give an example of such a case.

Yes. At the highest point of a vertically thrown ball, instantaneous velocity is zero, but gravitational acceleration remains about 10 m s^{-2} downward. Zero velocity at one instant does not imply zero rate of change of velocity.

D. Comprehensive Questions

2.1. How a vector can be represented graphically? Explain.

A vector is represented by a **directed line segment (arrow)**. Choose a suitable scale so that the line's length represents magnitude; the arrowhead and a stated angle or compass direction represent direction.

1. Draw suitable perpendicular reference axes and mark the positive directions.

2. Select a scale, for example 1 cm = 100 m s^{-1}.

3. From the origin, draw the calculated length at the stated angle, measured anticlockwise from the positive x-axis unless another convention is specified.

4. Add the arrowhead and label the vector with magnitude and unit.

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In symbols, bold letters such as **v** or an arrow over v distinguish a vector from its magnitude v.

2.2. Differentiate between: (i) rest and motion (ii) speed and velocity

Rest	Motion
Position does not change with time relative to a reference point.	Position changes with time relative to a reference point.
A parked car is at rest relative to the road.	The same car is moving relative to the road when driven.

Speed	Velocity
Scalar: magnitude only.	Vector: magnitude and direction.
Distance ÷ time.	Displacement ÷ time.
Always nonnegative.	May be positive or negative under a sign convention.
Example: 5 m s^{-1}.	Example: 5 m s^{-1} east.

2.3. Describe different types of motion. Also give examples.

Type	Description	Examples
Rectilinear translation	Every particle moves through the same straight-line displacement.	A car on a straight road; a freely falling stone.
Curvilinear translation	The body translates along a curved path without required rotation.	A car on a curved road; a thrown ball’s centre of mass.
Rotatory motion	A body turns about an axis; different particles move in circles about it.	Ceiling fan, wheel, Earth rotating on its axis.
Vibratory / oscillatory	Repeated motion to and fro about a mean position.	Pendulum, guitar string, mass on a spring.

Motion may also be uniform (equal displacements in equal times) or non-uniform (unequal displacements in equal times), and several types can occur together, as in a rolling wheel.

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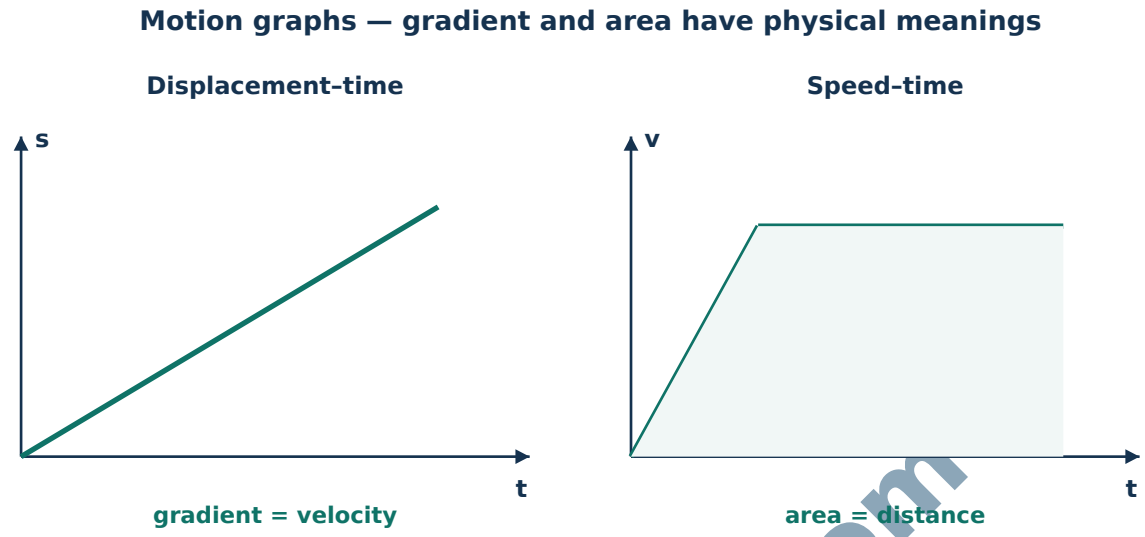
2.4. Explain the difference between distance and displacement.

Distance	Displacement
Total length of the actual path travelled.	Shortest directed change from initial to final position.
Scalar; no direction.	Vector; has direction.
Always positive for a completed journey.	Can be positive, negative or zero.
Depends on the route.	Depends only on initial and final positions.
Example: walking 60 m east then 60 m west gives 120 m.	The same journey gives zero displacement.

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2.5. What do gradients of distance-time graph and speed-time graph represent? Explain it by drawing diagrams.

Figure 2 — Gradient and area on motion graphs



Distance–time: gradient = Δdistance/Δtime = speed. A steeper line means greater speed; a horizontal line means rest. For a displacement–time graph, the corresponding gradient is velocity.

Speed–time: gradient = Δspeed/Δtime = acceleration. Positive gradient means speeding up, negative gradient means slowing down and zero gradient means constant speed.

2.6. Prove that the area under speed-time graph is equal to the distance covered by an object.

Divide the time interval into very small intervals Δt. If speed is approximately v during one interval, distance travelled in it is ΔS = vΔt.

Area of one narrow rectangle = height × width = v × Δt = ΔS

Total distance S = ΣΔS = Σ(vΔt) = total area under the speed–time graph

For constant speed, this is one rectangle S = vt. For uniform acceleration from rest, the graph is a triangle and S = ½ × time × final speed, agreeing with S = ½at² when v = at. For signed velocity, signed area gives displacement; area under the nonnegative speed graph gives distance.

2.7. How equations of motion can be applied to the bodies moving under the action of gravity?

Near Earth’s surface and neglecting air resistance, gravitational acceleration is approximately uniform, so the constant-acceleration equations apply with a replaced by g.

• **Choose a sign convention:** If upward is positive, use a = −g; if downward is positive, use a = +g.

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• **Dropped or freely falling body:** initial velocity u = 0 in the chosen frame.

• **Thrown upward:** velocity decreases to zero at the highest point and then reverses direction.

• **Use consistently:** $v = u \pm gt$, $S = ut \pm \frac{1}{2}gt^2$ and $v^2 - u^2 = \pm 2gS$, with signs determined by the convention.

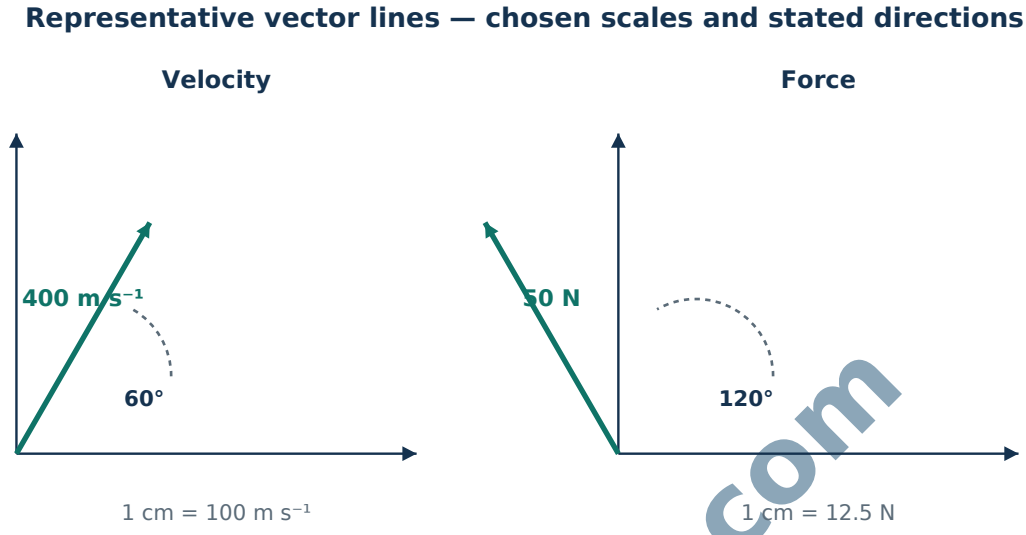
• **Use g ≈ 10 m s⁻²** for textbook calculations unless another value is supplied; displacement S is positive in the chosen direction, not automatically the total distance.

E. Numerical Problems

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2.1. Draw the representative lines of the following vectors. (a) A velocity of 400 m s⁻¹ making an angle of 60° with x-axis. (b) A force of 50 N making an angle of 120° with x-axis.

Figure 3 — Vector representations; different convenient scales are stated for the two quantities



• For velocity, a scale of 1 cm = 100 m s⁻¹ requires a 4.0 cm arrow at 60° anticlockwise from +x.

• For force, a scale of 1 cm = 12.5 N requires a 4.0 cm arrow at 120° anticlockwise from +x, i.e. 30° above the negative x-axis.

2.2 A car is moving with an average speed of 72 km h⁻¹. How much time will it take to cover a distance of 360 km?

Given: v = 72 km h⁻¹; S = 360 km.

Time = distance ÷ speed = 360 km ÷ 72 km h⁻¹ = 5 h

Answer: 5 h.

2.3 A truck starts from rest. It reaches a velocity of 90 km h⁻¹ in 50 seconds. Find its average acceleration.

Given: u = 0; v = 90 km h⁻¹ = 25 m s⁻¹; t = 50 s.

90 km h⁻¹ = 90 ÷ 3.6 = 25 m s⁻¹

$a = (v - u)/t = (25 - 0)/50 = 0.5 \text{ m s}^{-2}$

Answer: 0.5 m s⁻².

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2.4 A car passes a green traffic signal while moving with a velocity of 5 m s^{-1}. It then accelerates at 1.5 m s^{-2}. What is the velocity of car after 5 seconds?

Given: u = 5 m s^{-1}; a = 1.5 m s^{-2}; t = 5 s.

$$v = u + at = 5 + 1.5 \times 5 = 5 + 7.5 = 12.5 \text{ m s}^{-1}$$

Answer: 12.5 m s^{-1}.

2.5 A motorcycle initially travelling at 18 km h^{-1} accelerates at constant rate of 2 m s^{-2}. How far will the motorcycle go in 10 seconds?

Given: u = 18 km h^{-1} = 5 m s^{-1}; a = 2 m s^{-2}; t = 10 s.

$$S = ut + \tfrac{1}{2}at^2$$

$$S = 5 \times 10 + \tfrac{1}{2} \times 2 \times 10^2 = 50 + 100 = 150 \text{ m}$$

Answer: 150 m.

2.6 A wagon is moving on the road with a velocity of 54 km h^{-1}. Brakes are applied suddenly. The wagon covers a distance of 25 m before stopping. Determine the acceleration of the wagon.

Given: u = 54 km h^{-1} = 15 m s^{-1}; v = 0; S = 25 m.

$$v^2 - u^2 = 2aS$$

$$a = (0^2 - 15^2)/(2 \times 25) = -225/50 = -4.5 \text{ m s}^{-2}$$

Answer: −4.5 m s^{-2} (retardation of 4.5 m s^{-2}).

2.7 A stone is dropped from a height of 45 m. How long will it take to reach the ground? What will be its velocity just before hitting the ground?

Given: u = 0; S = 45 m downward; g = 10 m s^{-2}.

$$S = \tfrac{1}{2}gt^2 \rightarrow t = \sqrt{(2S/g)} = \sqrt{(90/10)} = 3 \text{ s}$$

$$v = u + gt = 0 + 10 \times 3 = 30 \text{ m s}^{-1} \text{ downward}$$

Answer: Time = 3 s; impact velocity = 30 m s^{-1} downward.

2.8 A car travels 10 km with an average velocity of 20 m s^{-1}. Then it travels in the same direction through a diversion at an average velocity of 4 m s^{-1} for the next 0.8 km. Determine the average velocity of the car for the total journey.

Given: S_1 = 10,000 m; S_2 = 800 m; same direction, so displacement = 10,800 m.

t_1 = 10,000/20 = 500 s; t_2 = 800/4 = 200 s

Average velocity = total displacement ÷ total time

= 10,800/(500 + 200) = 10,800/700 = 15.43 m s^{-1}

Answer: 15.4 m s^{-1} in the direction of travel.

2.9 A ball is dropped from the top of a tower. The ball reaches the ground in 5 seconds. Find the height of the tower and the velocity of the ball with which it strikes the ground.

Given: u = 0; t = 5 s; g = 10 m s^{-2}.

$h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 5^2 = 125$ m

$v = u + gt = 0 + 10 \times 5 = 50$ m s^{-1} downward

Answer: Height = 125 m; impact velocity = 50 m s^{-1} downward.

2.10 A cricket ball is hit so that it travels straight up in the air. It took 3 seconds to reach the highest point. What was the initial velocity of the ball? If the ball was hit 1 m above the ground, how high did it rise from the ground?

Given: At the highest point v = 0; t = 3 s; g = 10 m s^{-2} downward.

$0 = u - gt \rightarrow u = gt = 10 \times 3 = 30$ m s^{-1} upward

Rise above launch point = $u^2/(2g) = 30^2/(2 \times 10) = 45$ m

Maximum height above ground = 45 + 1 = 46 m

Answer: Initial velocity = 30 m s^{-1} upward; maximum height = 46 m above ground.

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F. More Important Questions (from the text)

1. How can average speed and average velocity differ? Give a numerical example.

Average speed = total distance ÷ total time, while average velocity = net displacement ÷ total time.

Suppose a student walks 60 m east in 30 s and then 60 m west in 20 s. Total distance = 120 m, total time = 50 s, so average speed = 120/50 = **2.4 m s^{-1}**. Net displacement is zero, so **average velocity = 0**.

2. How do curved and horizontal motion graphs help describe a journey?

• On a distance–time graph, an increasing gradient means increasing speed, a decreasing gradient means decreasing speed and a horizontal segment means rest.

• On a displacement–time graph, a negative gradient means motion in the negative direction; a curved line means changing velocity.

• On a speed–time graph, a horizontal line means constant speed, a rising line means positive acceleration and a falling line means retardation.

• The area under a speed–time graph gives distance; the area under a signed velocity–time graph gives displacement with direction.

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Points to Remember

Distance is path length; **displacement** is directed change in position.

Average speed = total distance/time; **average velocity** = displacement/time.

Acceleration = change in velocity/time; direction change alone can produce acceleration.

Gradient: distance–time → speed; displacement–time → velocity; speed–time → acceleration.

Area: under speed–time → distance; signed area under velocity–time → displacement.

Equations: $v = u + at$; $S = ut + \frac{1}{2}at^2$; $v^2 - u^2 = 2aS$.

Conversion: km $h^{-1} \div 3.6$ = m s^{-1}; m $s^{-1} \times 3.6$ = km h^{-1}.

Free fall: from rest u = 0 and distance in first second = 5 m when g = 10 m s^{-2}.

Thrown upward: at the top velocity is zero but acceleration remains g downward.

Vector drawing: length uses a scale; arrowhead and stated angle give direction.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	d	2	a	3	a	4	b	5	b
6	a	7	d	8	c	9	a	10	c

How to attempt this chapter in the paper

- Convert km h^{-1} to m s^{-1} before using equations with seconds and metres.
- Write u, v, a, S and t with signs before selecting an equation; a negative acceleration may represent retardation or direction.
- For graph questions, read two coordinates and calculate gradient explicitly; for area, state the geometric shape.
- For average velocity over two stages, add displacements and times—not the two velocities.
- For vertical motion, state the positive direction and distinguish maximum height above launch from height above ground.